

2501-PYL1003 (Mechanics & Special Theory of Relativity)

Practice Mid-Semester Examination — Paper 3 (Unofficial, Instructor-Style)

Full Marks: 60 Time: 2 hours Set: Practice-3

Instructions

An MCQ can also be an MSQ (select all correct options). Justify every choice — no marks for a bare letter. Show all intermediate steps and reasoning.

State any approximation or rounding explicitly, with justification.

Numerical answers should be rounded to two decimal places unless stated otherwise.

1. A damped oscillator's displacement is empirically modelled as

$$x(t) = A e^{-\beta t} \cos(\omega t + \delta)$$

where x has dimension of length and t of time. (i) State the dimensions of A , β , ω , and δ , justifying each from the requirement that the exponent and the cosine argument must be dimensionless. (ii) The "quality factor" of such an oscillator is often defined as $Q = \omega / (2\beta)$. Show that Q is dimensionless. **[4+2 = 6]**

2 (MSQ). Planck units are constructed from G , $\hbar$, c alone. Which of the following combinations has dimension of **length**? (Justify by dimensional analysis; do not just recall the numerical Planck length formula.) a) $\sqrt{\hbar G / c^3}$ b) $\sqrt{\hbar G / c^2}$ c) $\hbar / (Gc)$ d) $G\hbar / c^4$ **[5]**

3. A particle moves along the x -axis with velocity given as a function of position: $v(x) = v_0 e^{-x/L}$ for $x \geq 0$, where v_0 , L are positive constants and the particle starts at $x = 0$ at $t = 0$. (i) Find $x(t)$ explicitly. (ii) Find the acceleration $a(x)$ as a function of x using $a = v dv/dx$, and confirm it is consistent with differentiating your $x(t)$ twice. (iii) Does the particle reach $x = L \ln 10$ in **finite** time? Justify quantitatively (i.e., compute the time, or show the relevant integral diverges/converges). **[3+3+3 = 9]**

4. A block of mass 2 kg on a frictionless horizontal surface is attached to a wall by a spring of natural length 0.5 m and force constant $k = 50$ N/m. The block is pulled so the spring stretches to length 0.9 m and released from rest. Simultaneously (from the moment of release), a constant horizontal force of 4 N acts on the block in the direction that **opposes** the initial spring pull-back (i.e., aiding the spring's restoring pull initially). (i) Write the equation of motion and identify the new equilibrium position (measured as spring extension from natural length). (ii) Find the amplitude of the resulting oscillation about this new equilibrium. (iii) Find the maximum speed attained by the block. **[2+3+2 = 7]**

5 (MSQ). A particle's position vector in spherical polar coordinates has constant $r = 4$ and is confined to move only in θ and ϕ . At one instant, $\theta = 60^\circ$, $\phi = 30^\circ$, $\dot{\theta} = 2$ rad/sec, $\dot{\phi} = 1$ rad/sec. Recall the velocity in spherical polar coordinates: $\vec{v} = \dot{r} \hat{r} + r\dot{\theta} \hat{\theta} + r \sin \theta \dot{\phi} \hat{\phi}$. Which of the following is/are correct about the speed $|\vec{v}|$ at this instant (two decimal places)? a) The $\hat{\theta}$ -component of velocity is 8.00 m/sec b) The $\hat{\phi}$ -component of velocity is 3.46 m/sec c) The speed is 8.72 m/sec d) The speed is 8.00 m/sec (since $\dot{r} = 0$, only the $\hat{\theta}$ term contributes) **[6]**

6. Using the standard relations $\hat{\rho} = \cos \phi \hat{i} + \sin \phi \hat{j}$, $\hat{\phi} = -\sin \phi \hat{i} + \cos \phi \hat{j}$, $\hat{z} = \hat{k}$ for cylindrical coordinates: (i) Derive $\frac{d\hat{\rho}}{dt}$ and $\frac{d\hat{\phi}}{dt}$ in terms of $\dot{\phi}$ and the unit vectors themselves (i.e., show $\dot{\hat{\rho}} = \dot{\phi} \hat{\phi}$ and $\dot{\hat{\phi}} = -\dot{\phi} \hat{\rho}$), by direct differentiation with respect to time. (ii) Hence derive the standard expression for acceleration in cylindrical coordinates, $\vec{a} = (\ddot{\rho} - \rho\dot{\phi}^2)\hat{\rho} + (\rho\ddot{\phi} + 2\dot{\rho}\dot{\phi})\hat{\phi} + \ddot{z}\hat{z}$, starting from $\vec{r} = \rho\hat{\rho} + z\hat{z}$ and differentiating twice. **[4+5 = 9]**

7. A binary star system consists of two stars in circular orbits about their common center of mass, separated by a fixed distance $d = 2 \times 10^{11}$ m. Star 1 has mass $M_1 = 3 \times 10^{30}$ kg and Star 2 has mass $M_2 = 1 \times 10^{30}$ kg ($G = 6.674 \times 10^{-11}$ N m²/kg²). (i) Find the distance of each star from the center of mass. (ii) Using the fact that the mutual gravitational force provides the centripetal force for the **relative** orbit (reduced two-body problem, effective separation d , total mass $M_1 + M_2$), find the orbital period of the system. **[3+4 = 7]**

8. A particle moves under a central force $F(r) = -\frac{k}{r^n}$ for general power n . A circular orbit of radius R exists for any n (given suitable L). (i) Derive the general **stability condition** on n for small radial perturbations about the circular orbit to remain bounded (oscillatory), in terms of n alone. (You should arrive at a condition like $n < n_{\text{crit}}$ for some critical value.) (ii) Check your condition against the two special cases $n = 2$ (inverse-square) and $n = 3$ (inverse-cube, as in Paper 1 Q15) and comment on consistency. **[6+3 = 9]**

9. A river flows due south at a constant speed $u = 2$ m/sec relative to the ground. A boat travels due east relative to the water at constant speed $w = 5$ m/sec, on Earth at latitude $\lambda = 60^\circ$ N. Consider the boat's motion over a duration of $\tau = 3600$ sec, and estimate the **additional** deflection caused by the Coriolis effect (relative to the naive straight-line-relative-to-water path), given Earth's $\Omega = 7.29 \times 10^{-5}$ rad/sec. State which horizontal direction (further north/south of the naive path) the Coriolis correction pushes the boat, with reasoning, and estimate its magnitude using the zeroth-order (no-Coriolis) velocity in the correction term. **[7]**

10. A bead of mass m slides without friction on a **rigid circular hoop** of radius b that is forced to rotate at constant angular velocity Ω about a **vertical diameter** of the hoop (so the hoop spins about a vertical axis through its center, and the bead's position on the hoop can be described by the angle α from the bottom of the hoop, with gravity acting downward). (i) Working in the frame rotating with the hoop, write the tangential (along-the-hoop) equation of motion for $\alpha(t)$, including gravity and the centrifugal term (the Coriolis term has no tangential component for motion confined to the hoop — you may state this without proof). (ii) Find all equilibrium angles α_{eq} (where $\dot{\alpha} = \ddot{\alpha} = 0$) in terms of g, b, Ω , and determine the condition on Ω (in terms of g, b) for a nonzero equilibrium $\alpha_{eq} \neq 0, \pi$ to exist. **[3+5 = 8]**

11. A two-stage rocket in free space (no gravity, no drag) has total initial mass M_0 . Stage 1 has fuel mass m_1 and, when burned completely (at constant exhaust speed u relative to the rocket), is then **jettisoned** (its now-empty casing, mass m_{c1} , separates and is left behind, carrying no further fuel). Stage 2 then burns its fuel mass m_2 (same exhaust speed u) to reach final structural mass m_f . (i) Using the rocket equation stage-by-stage, write the total final speed v_f (starting from rest) as a sum of two logarithmic terms. (ii) Given $M_0 = 100$ tonnes, $m_1 = 60$ tonnes fuel with casing $m_{c1} = 10$ tonnes (jettisoned after stage 1), stage 2 total mass before its burn = 30 tonnes of which $m_2 = 24$ tonnes is fuel, $u = 3000$ m/sec: compute v_f , and compare it (briefly, qualitatively) to the speed a **single-stage** rocket of the same total initial and final mass would achieve — is staging advantageous here, and why (conceptually, no need to fully recompute the single-stage case unless you wish to verify)? **[3+5 = 8]**

12. A rope of total mass m and length L hangs vertically, with its lower end just touching a table. The rope is released from rest at $t = 0$ so that it falls freely under gravity (the top end is not held — the entire rope is in free fall as a rigid unmoving-relative-to-itself column until contact). As the rope piles up on the table, each element that lands comes instantaneously to rest (fully inelastic; the pile does not rebound). (i) At the instant when a length x of rope has landed on the table (with $0 < x < L$), find the speed of the still-falling part of the rope (it is in free fall throughout, so this is straightforward kinematics). (ii) Find the **normal force** exerted by the table on the pile at that instant. (Hint: this force must support the weight of the already-landed pile **and** provide the impulsive force to stop the incoming rope — compute both contributions using the variable-mass framework, i.e., $F_{\text{table}} - (\text{weight of pile}) = \frac{d}{dt}(\text{momentum absorbed})$, taking care with signs.) Express your answer in terms of m, L, g, x . **[3+6 = 9]**

13. A crate of mass $M = 40$ kg rests on a rough horizontal floor ($\mu_s = 0.5, \mu_k = 0.4$). A rope is attached to the crate and passes over a small frictionless pulley fixed at height h above the floor, then down to a hanging counterweight of mass m . The rope from the crate to the pulley makes an angle $\theta = 30^\circ$ with the horizontal (i.e., the crate is pulled at an angle, not just horizontally) because the pulley is elevated. (i) Draw the free-body diagram of the crate (describe in words: forces and their directions) and write the horizontal and vertical equilibrium equations for the **crate**, treating the tension T (pulling at 30° above horizontal toward the pulley) and normal force N as unknowns, for the case where the system is on the verge of sliding. (ii) Find the minimum hanging mass m required to just start moving the crate ($g = 10$ m/sec²). **[3+5 = 8]**

14 (MSQ). A uniform ladder of length ℓ and mass m leans against a smooth (frictionless) vertical wall, with its foot on a rough horizontal floor (coefficient of static friction μ_s), making angle θ with the floor. The ladder is on the verge of slipping. Which of the following statement(s) is/are correct (you may use torque balance about the foot of the ladder, and force balance; take the ladder's weight to act at its midpoint)?
a) The condition for the ladder not to slip is $\tan \theta \geq \frac{1}{2\mu_s}$ b) The condition for the ladder not to slip is $\tan \theta \geq \frac{1}{\mu_s}$ c) The normal force from the wall equals the friction force from the floor at the verge of slipping d) Increasing μ_s allows the ladder to be stable at **smaller** angles θ (closer to horizontal) **[6]**

15. Consider a particle of mass m constrained to move on the inside surface of a frictionless vertical cone with half-angle α (apex pointing down, axis vertical), i.e., the particle's cylindrical radius ρ and height z (measured from the apex) satisfy $\rho = z \tan \alpha$. Under gravity, the particle can execute motion with both a

"height" degree of freedom and an angular degree of freedom ϕ about the axis. Using cylindrical coordinates and eliminating z in favor of ρ via the constraint: (i) Write Newton's second law in the $\hat{\rho}$ direction (radial, i.e., along the surface's slant), including the normal force component and gravity's component along the surface, plus the centrifugal-type term $-\rho\dot{\phi}^2$ appearing from the curvature of the path (this stays in the *inertial*-frame Newton's law since ϕ genuinely changes; do not confuse with a rotating-frame problem). (ii) Show that a **horizontal circular orbit** at fixed $\rho = \rho_0$ (i.e., $\dot{\rho} = \ddot{\rho} = 0$) is possible, and find the required $\dot{\phi}$ in terms of g, ρ_0, α . **[4+5 = 9]**

End of Paper 3. Total: 60 marks.